

Yuxuan Liu

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Research interests: Model reinforcement learning; inference optimization; trajectory analysis



🎓 Education

Tsinghua University | Dual Degree Sep. 2023 – Jul. 2027 (expected)
B.Eng. in Computer Science and Technology; B.Econ. in Economics and Finance GPA: 3.3 / 4.0
Coursework: Software Engineering, Computer Graphics, Game Theory, SRT, Machine Learning

⚙️ Publications & Research Experience

Boosting Fine-Grained Visual Perception via Multi Granularity Data Synthesis and Local-to-Global Curriculum Learning

W. Yan#, Y. Liu#, D. Zou et al. | **NAACL 2027**, under review Feb. 2026 – Present
Led data synthesis and experiments for local-to-global curriculum RL to improve fine-grained VLM perception; co-developed the method and co-wrote the paper.

From Cache to Belief: Online Probabilistic Memory for Long-Context Language Models

J. Wu, Y. Liu & Y. Yue* et al. | **NeurIPS 2026**, under review Mar. 2026 – Aug. 2026
Designed the evaluation protocol and ran system experiments for BeliefMem, an online Bayesian memory adapter for frozen language models; analyzed results and co-wrote the paper.

You Only Show Once: Robust GUI Automation via One-Shot Human Demonstration

Y. Huang#, Y. Liu#, W. Yan et al. | **NAACL 2027**, under review Sep. 2025 – Mar. 2026
Contributed to problem formulation, data processing and experiments for demonstration-guided GUI agents with visual grounding and reflection-based recovery.

A Scalable Benchmark Test Suite for Dynamic Multi-Objective Optimization with a Changing Number of Objectives

K. Shang, Z. Xiao, Y. Liu et al. | **PPSN 2026**, accepted Aug. 2025 – Dec. 2025
Built scalable benchmarks that vary the number of active objectives while keeping objective functions fixed; designed and ran comparative evaluations.

A Device-Cloud Collaborative Multimodal Inference Method and System

Y. Liu | Chinese invention patent, under substantive examination App. No. 2026105681825
Designed dynamic allocation of perception and reasoning across device and cloud to reduce on-device inference costs and strengthen privacy protection.

Graph-Generating LLMs for Drug Discovery | SRT, Dept. of CST

Jun. 2024 – Jun. 2025
Built data and prompt synthesis pipelines and post-trained DeepSeek-R1 and other LLMs with RL; compared models and proposed a benchmark for rule-constrained graph generation.

👛 Internships

ZTE Corporation | AI Research Intern Aug. 2025 – Mar. 2026
Researched multimodal LLM inference acceleration and visual perception; independently built data synthesis and evaluation pipelines for fine-tuning and ablation studies.

Meituan | World Model Intern, Autonomous Vehicles Jun. 2026 – Jul. 2026
Processed multimodal data and trained autonomous-driving world models; evaluated and optimized perception for robustness in complex environments.

👥 Service & Leadership

Student Association for Science and Technology, Dept. of CST Jun. 2025 – Present
Vice President (AI Agent Division). Led development of the 29th AI Agent Contest' s official game player and visualizer; coordinated university-wide contests, problem design and technical support.
Led communications and teaching activities during research fieldwork in Japan and volunteer teaching in Qinghai.

★ Honors & Awards

Outstanding Social Work Scholarship, Tsinghua University 2025
Third Prize, Youth Digital Innovation Competition (Beijing Xicheng) 2025
Winning Prize, 7th “Bambu Lab Cup” Software Design Competition, Dept. of EE 2024

🔧 Skills

PyTorch, Transformers; LLM training, fine-tuning and evaluation; multi-GPU distributed training and experiment management on Linux; full-stack development, Git collaboration and code review.